

3.5 Comparison Tests Worksheet

Directions: Determine if each of the following series converges or diverges. Use any technique from section 3.1 to section 3.5.

1.
$$\sum_{n=1}^{\infty} \frac{n}{2n^3 + 1}$$

2.
$$\sum_{n=1}^{\infty} \frac{1}{2n+3}$$

3.
$$\sum_{k=1}^{\infty} \frac{1}{(3+k)^{2/5}}$$

4. $\sum_{k=1}^{\infty} \frac{\cos\left(\frac{1}{k}\right)}{k^2}$

5. In the series $\sum_{n=1}^{\infty} \left(\frac{2^n}{x^n}\right)$, x stands for a real number. Depending on what number x is replaced with the given series may converge or diverge. Write an argument to find the values of x for which $\sum_{n=1}^{\infty} \left(\frac{2^n}{x^n}\right)$ converges.

6. $\sum_{k=1}^{\infty} \left(\frac{3}{2^k} - \frac{2}{3^k}\right)$

$$7. \sum_{k=1}^{\infty} \left(2 - \frac{1}{k} \right)$$

$$8. \sum_{n=1}^{\infty} \frac{4^{n+1}}{3^n - 2}$$

$$9. \sum_{n=1}^{\infty} \frac{n+2}{(n+1)^3}$$

$$10. \sum_{k=1}^{\infty} \frac{\arctan(k)}{1+k^2}$$

$$11. \sum_{n=1}^{\infty} \frac{n^2 - 5n}{n^3 + n + 1}$$